M13-C3 — Probe catalogue (H1–H4) motivated by closed H0

H0 closed; H1–H3 run/analysed;
H4 proposed / infra-blocked (not claims)

1 Status

Follow-up probes **motivated by closed baseline H0**, and **not** part of H0 itself. H0 is the completed six-cell AAMAS+ECAI investigation on fixture `m13_bucket3_binary_collider_and` / `n30_seed42`, documented in `learning_analysis.md` / `.tex` and summarised in `experiment.md`.

- H0: **closed** (see `learning_analysis.md`; not redefined here)
- H1: **run / analysed** — nested-prefix AAMAS ablation on `n25_seed42`. Record: `h1_support_ablation.md` / `.tex`. **Not a Bucket 3 claim.**
- H2: **run / analysed** — isolated-D AAMAS target c retains irrelevant d. Record: `h2_irrelevant_covariate.md` / `.tex`. **Not a Bucket 3 claim.**
- H3: **run / analysed** — BK-leading isolated a distracts ECAI target c from BD AND. Record: `h3_bk_leading_distractor.md` / `.tex`. **Not a Bucket 3 claim.**
- H4: **proposed** / blocked on target-wise cautious support / not tested / not a Bucket 3 claim
- **Not** approved Bucket 3 claims
- **Not** locked Bucket 1/2 claim content
- **Not** report-facing prose (`docs/report/findings/` is out of scope)
- Markdown twin: `future_probes.md`

Samuel decision (2026-08-01) — H1 / H2.

1. **H1 approved** — nested-prefix / missing-$(0, 0, 0)$ AAMAS ablation on this fixture (e.g. $n = 25$, seed 42, targets a and b) against the approved $n = 30$ baseline.
2. **H2 design approved** — separate schema-2 fixture with isolated $D \sim$ Bernoulli(1/2) (variables `a`,`b`,`c`,`d`).
3. **H2c in scope** for the first H2 write-up.

Samuel decision (2026-08-01) — H3.

1. Graph/naming sketch **approved**.
2. Root parameters **approved:** $B \sim$ Bernoulli(4/5) ($P(B{=}1) = 4/5$, $P(B{=}0) = 1/5$); $D \sim$ Bernoulli(7/10) ($P(D{=}1) = 7/10$, $P(D{=}0) = 3/10$); $A \sim$ Bernoulli(1/2) as previously sketched.
3. Fixture id **locked:** `m13_bucket3_binary_bd_and_lead_a` (AND of parents `b`,`d`; BK-leading isolated `a`). Distinct from `m13_bucket3_binary_collider_and` and from the completed H2 fixture id `m13_bucket3_binary_collider_and_iso_d`.

4. **AAMAS contrast in scope** for the H3 write-up (secondary to primary ECAI inspection of target c; valuable as a related but distinct check from H2).

H4. Signposted only; **not** approved for implementation and **blocked** until target-wise cautious learning is supported (see H4 section).

H1–H3 have been run and analysed (`h1_support_ablation.md`, `h2_irrelevant_covariate.md`, `h3_bk_leading_distractor.md`). Approvals do not create a Bucket 3 claim or a run matrix.

2 Motivation (verified facts from the AAMAS arm of closed H0)

On the approved frozen sample `n30_seed42`, configuration `aamas2025` (greedy / `folding_selection(mgr)` / `folding_space(bk)` / brave / `check_ic`):

Target	Outcome	Delta (if any)
`c`	`solved`	`c(A) :- a_val_1(A), b_val_1(A).` (assumption-free)
`a`	`completed_no_solution`	empty
`b`	`completed_no_solution`	empty

Sources:

- collection summary: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/summary.md`
- target c delta: `.../cells/target-c/output/delta.aba`
- target a/b reports and `prolog.stdout` under `.../cells/target-{a,b}/`

For target c, the learned delta string coincides with the evaluator-only positive-state reference in `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/mechanism_reference.json`. Record that as **syntactic coincidence**, not as a causal-recovery verdict.

Procedural reading of the AAMAS root-target path (verified in `.../cells/target-a/output/prolog.stdout`; target b is symmetric):

1. Greedy folding builds **maximal co-occurrence bodies** from BK for each E^+ row (not a minimal-parent search).
2. The both-1 co-occurrence body `a(A) :- b_val_1(A), c_val_1(A)` is accepted under brave entailment.
3. The both-0 co-occurrence body `a(A) :- b_val_0(A), c_val_0(A)` over-generalises; assumption/contrary repair is attempted.
4. Contrary rote learning cites rows **26** and **30** — the only sample atoms $(0, 0, 0)$ in `samples/n30_seed42.csv` — and contrary folding then fails, yielding `completed_no_solution`.

These facts from closed H0 motivate probe families H1 and H2. They do not by themselves approve a Bucket 3 claim.

2.1 Motivation addendum (ECAI arm of closed H0 — for H3)

On the same frozen sample `n30_seed42`, configuration `ecai2024` (`folding_mode(nd)`, `folding_selection(any)`, brave / `check_ic`), target c **solved** with delta (verified):

```
c(A) :- alpha_1(A), a_val_1(A).
```

```
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

In `.../ecai2024/n30_seed42/cells/target-c/output/prolog.stdout`, the first nd fold on an early E^+ row replaces the sample id by `a_val_1(A)` (the first BK predictor under declared order `a`, `b` for that cell), then assumption repair retains a in the ordinary target body while the second parent enters via a contrary. Collection summary: `.../ecai2024/n30_seed42/summary.md`.

This motivates **H3**: put an *independent* distractor first in BK by naming, under ECAI, on a new fixture. H3 is now **run / analysed** (`h3_bk_leading_distractor.md`); it is still **not** a Bucket 3 claim.

3 Ordering

1. **H1** — nested-prefix / missing-$(0, 0, 0)$ ablation on the existing fixture and AAMAS config (seed 42). **Run / analysed** (`h1_support_ablation.md`).
2. **H2** — separate schema-2 fixture with isolated covariate D on the $A \to C \leftarrow B$ AND collider. **Run / analysed** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains `d_val_*`.
3. **H3** — separate schema-2 fixture `m13_bucket3_binary_bd_and_lead_a` with BK-leading independent A and AND collider $B \to C \leftarrow D$. **Run / analysed** (`h3_bk_leading_distractor.md`). Lead: ECAI first-folds to a and retains a in the theory for c.
4. **H4 (proposed)** — same fixture/`n30_seed42`; ECAI **root** targets under **cautious** vs locked brave; **blocked** on target-wise cautious support. Not approved for run until infra is designed and approved.

Do not expand any family into a broader run matrix beyond these approved probes (and a later-approved H4) without a further Samuel decision.

3.1 Relation among probe families (keep distinct)

Probe	Focus
H1	AAMAS roots; missing negative witness / both-0 over-generalisation on the **current** three-variable fixture
H2	AAMAS target c; irrelevant covariate enters **greedy maximal** co-occurrence bodies (minimality vs solvability); fixture id `m13_bucket3_binary_collider_and_iso_d`
H3	**ECAI** target c (primary); **BK order** puts independent distractor **first**; nd first-fold / repair path; **AAMAS contrast in scope**; id `m13_bucket3_binary_bd_and_lead_a`
H4	**Same fixture/sample**; ECAI **root** targets a/b; **brave** $\to$ **cautious**; **requires target-wise learning-mode support**

H3 is not a duplicate of H2: different DAG, different distractor placement/name, different primary config (ECAI nd vs AAMAS greedy), and distinct fixture ids. The in-scope AAMAS arm on H3 still differs from H2 because the irrelevant variable is BK-leading a and the mechanism parents are b, d.

H4 is not a duplicate of H1 or H3: it holds the table fixed and varies **entailment / learning mode** on the target-wise path for **roots**.

4 Probe family H1 — support sensitivity on AAMAS root targets

Status: **run** / **analysed** / not a claim. Evidence record: `h1_support_ablation.md` / `h1_support_ablation.tex`.

Evaluative stance (locked for H1 reading). Roots have `no_observed_parent_deterministic_rule`. H0 AAMAS root `completed_no_solution` is the **desired** / **correct** outcome. H1 AAMAS root `solved` with co-occurrence Horn rules is an **incorrect** / **spurious** finite-sample solution.

4.1 H1a — spurious root rule under missing negative witness

On this fixture, AAMAS, target a (symmetrically b), seed 42: the nested prefix n=25 **omits all** $(0, 0, 0)$ **rows**. Learning **solved** with a delta that includes the both-0 co-occurrence rule

```
a(A) :- b_val_0(A), c_val_0(A).
```

(respectively `b(A) :- a_val_0(A), c_val_0(A).`), which is **false** as a population implication and is blocked on the closed H0 n=30 table by E^- rows 26 and 30.

Nested-prefix fact (seed 42): rows 1–25 contain no $(0, 0, 0)$ atom; rows 26 and 30 are the only $(0, 0, 0)$ observations in n=30. Verified: `n25_seed42.csv` is a row-for-row prefix of `n30_seed42.csv`.

4.2 H1b — negative witness in the unsafe cell

If the sample contains at least one opposite-label row in that both-0 predictor cell, the both-0 rule fails brave entailment under the observed AAMAS path. On closed H0 n=30 this manifested as `completed_no_solution`, not as emission of only the safe both-1 rule. That H0 outcome is the **correct** reading for roots under the H1 stance.

4.3 Explicit non-claims for H1

- Full support does **not** imply AAMAS will emit a "correct" deterministic root mechanism rule: roots have `no_observed_parent_deterministic_rule` in the evaluator reference.
- "One of each population atom" is a sufficient practical condition on this approved sample, but the operational point is a **label conflict inside the greedy candidate body cell**, not a generic demand for full joint support.
- H1 does not claim that omitting $(0, 0, 0)$ is the only way AAMAS can emit a spurious root rule, nor that every seed behaves identically.
- H1 does **not** claim that root `solved` on n=25 is progress or mechanism recovery.

4.4 Outcome (H1)

Run / analysed. Nested-prefix ablation executed on `m13_bucket3_binary_collider_and` / AAMAS / seed 42 / n=25. Roots a/b **incorrectly** solved with both-1 and both-0 co-occurrence rules. Child c remained AND-solved as a control. Full narrative: `h1_support_ablation.md`.

5 Probe family H2 — irrelevant isolated covariate on AAMAS target c

Status: **run** / **analysed** / not a claim. Evidence record: `h2_irrelevant_covariate.md` / `h2_irrelevant_covariate.tex`.

Lead finding. On fixture `m13_bucket3_binary_collider_and_iso_d` / `n30_seed42`, AAMAS solves target c with

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

so **irrelevant D is included**. The evaluator-only reference remains the D-free AND `c(A) :- a_val_1(A), b_val_1(A).`. Failure mode: distractor retention / minimality, not unlearnability.

5.1 H2a — irrelevant covariate enters greedy bodies

Observed as predicted: when both values of D appear among E^+ for c (9 each on n=30), greedy catalogue bodies **include** `d_val_*`.

5.2 H2b — learnability vs minimality

c **still solved** (two D-conditioned rules). Confirms that the interesting failure is retention of an irrelevant predictor, not `completed_no_solution`.

5.3 H2c — sample-edge (also run; not the centre)

Nested n=2 prefix where C=1 realises only d=1: single rule `c :- a_val_1, b_val_1, d_val_1`. Extreme edge. Documented briefly in `h2_irrelevant_covariate.md`. Not the scientific centre of H2.

5.4 Outcome (H2)

Run / analysed. Fixture authored and certified; AAMAS `n30_seed42` primary target c shows distractor inclusion; H2c n=2 also run. Full record: `h2_irrelevant_covariate.md`.

6 Probe family H3 — ECAI first-fold distraction by BK-leading independent variable

Status: **run** / **analysed** / not a claim. Evidence record: `h3_bk_leading_distractor.md` / `h3_bk_leading_distractor.tex`.

Lead finding. On fixture `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42`, ECAI first-folds to `a_val_*` and retains a in both target rules for c. The emitted delta does **not** coincide with the evaluator-only BD AND `c(A) :- b_val_1(A), d_val_1(A).`. Failure mode: BK-leading distractor inclusion / first-fold distraction, not unlearnability.

6.1 Fixture (as run)

Field	Value
Fixture id	`m13_bucket3_binary_bd_and_lead_a`
Internal / display	`a`,`b`,`c`,`d` / A, B, C, D
DAG	$B \to C \leftarrow D$; **no edges** involving A
A	Bernoulli(1/2); independent of $\{B, C, D\}$
B	Bernoulli(4/5) — $P(B{=}0) = 1/5$, $P(B{=}1) = 4/5$
D	Bernoulli(7/10) — $P(D{=}0) = 3/10$, $P(D{=}1) = 7/10$
C	deterministic $C = B \wedge D$
Evaluator reference for c	`c(A) :- b_val_1(A), d_val_1(A).` (`a` must not appear)
BK order for target c	predictors `a`**, then** `b`**, then** `d` (irrelevant first)
Frozen sample	`n30_seed42`
Primary learning arm	**ECAI** target c
Secondary arm	**AAMAS** on the same frozen sample (**in scope**)

6.2 Scientific question (answered)

When learning target c under **ECAI** on this frozen sample, does BK-leading independent a **distract** the first fold and/or the final ABA shape away from the mechanism-aligned B, D conjunction?

Answer (bounded): yes — first fold selects `a_val_1`; both target rules retain a; metrics body vars $\{a\}$. Secondary: AAMAS also retains `a_val_*` in both Horn bodies (vs H2's trailing-d inclusion).

6.3 Secondary finding (in scope; not the lead)

Brave residual / underdetermining delta on the nested $a{=}1$ / α_1–α_3 side. Cross-link H0 ECAI target-a rows 9 vs 26. Reinforces: `solved` + **audit SAT** $\neq$ **mechanism-aligned**.

6.4 Explicit non-claims for H3

- Inclusion of a is **not** causal recovery of an $A \to C$ edge.
- Runner `solved` is **not** automatically correct / mechanism-aligned.
- H3 does not reuse H2's graph; distractor here is A, parents B, D.
- Completing H3 does **not** approve a Bucket 3 claim.

6.5 Outcome (H3)

Run / analysed. Fixture authored and certified; ECAI `n30_seed42` primary target c shows BK-leading a distraction; AAMAS contrast also retains a. Full record: `h3_bk_leading_distractor.md`.

7 Probe family H4 — cautious vs brave on ECAI root targets

Status: **proposed** / blocked on target-wise cautious support / not tested / not a Bucket 3 claim.

Uses the **existing** fixture and frozen sample `m13_bucket3_binary_collider_and` / `n30_seed42`. Does **not** require a new DAG. Cannot be executed with today's target-wise validator as-is.

7.1 Motivation (verified facts)

On ECAI **brave** (`configs/ecai2024_config.pl`: `learning_mode(brave)`), targets a and b **solved** with assumption-rich deltas (summary: 11 delta rules, 3 assumptions each). Artefacts:

- `.../ecai2024/n30_seed42/cells/target-a/`
  Files: `delta.aba`, `prolog.stdout`, and `bk.sol_chk.asp`
- mirror: `.../cells/target-b/`
- collection summary: `.../ecai2024/n30_seed42/summary.md`

Verified sample atoms (seed 42):

Row	(A, B, C)	Role for target a
9	$(1, 0, 0)$	E^+
26	$(0, 0, 0)$	E^-

Rows 9 and 26 share identical BK literals `b_val_0`, `c_val_0`. No deterministic rule over (B, C) assigns both labels. The brave delta for a includes the α_2/α_3 nest (verified in `delta.aba`):

```
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

Trace/fixture integration reading (for the signpost): this is a **per-row assumption choice gadget** — one stable model can make the target hold at row 9 and reject it at row 26 because ground assumption atoms differ. Runner `solved` means brave coverage of E^+/E^-, **not** recovery of a functional root mechanism (`no_observed_parent_deterministic_rule`).

Engine fact (`asp_engine.pl`, `entails/5`): brave entailment requires existence of a stable model under joint E^+/E^- constraints; cautious entailment requires E^+ atoms to lie in **cautious consequences** (true in every answer set) and E^- atoms not to. Under that reading, the current brave delta should **fail** cautious entailment: e.g. $a(9)$ is not true in every stable model of the choice gadget. Cautious re-check after contrary rote is invoked in `gen.pl` (`gen6`).

7.2 Infrastructure prerequisite (required before H4 can run; not yet done)

H4 is **not** "point YAML at a cautious `.pl` and run." The **target-wise** path currently **hard-requires brave**.

Verified code fact (`causal/targetwise/config.py`, `_read_learning_mode`): the runner reads `set_lopt(learning_mode(...))` from the Prolog config and **rejects** any mode other than `brave`, with an error that the current target-wise runner supports brave learning only. The target-wise README states that the supported regime selects brave learning. Existing configs only supply `learner.prolog_config` (e.g. `configs/ecai2024_config.pl`); learning mode is **not** a free YAML field — it is derived from the consulted `.pl` file and then validated.

Two-stage plan:

1. **Infrastructure / config design (prerequisite, careful, small)**
   Decide how cautious mode is exposed for target-wise without breaking existing brave collections or hashes. Likely pieces (for the record; final design needs Samuel/implementation approval):

- a **new** Prolog options file (do **not** edit `configs/ecai2024_config.pl` in place), e.g. an ECAI-cautious twin that keeps nd / any / all / relto / `check_ic` but sets `learning_mode(cautious)`;
- relax or extend `causal/targetwise/config.py` (and README/tests) so `learning_mode(cautious)` is an **explicitly allowed** target-wise mode when selected that way;
- a **new** `configuration.id` and output directory so brave `ecai2024` artefacts remain untouched;
- confirm what `check_ic` / `.sol_chk.asp` / the post-run artefact audit mean under cautious learning (joint existential SAT vs cautious consequences) so diagnostics are not misread.

Mode is coupled to Prolog consult, config hashing, manifests, and the brave-only validator. Do not imply that only a YAML one-liner is required.

2. **Scientific run (only after the above)**
   Target-wise collection on the existing fixture/`n30_seed42` for roots a/b (optional c control) under the new cautious configuration.

7.3 Precise hypotheses (not claims)

H4a (current delta is brave-specific). The serialized ECAI brave delta for target a on this sample does **not** cautiously entail the E^+/E^- set: in particular, both-0 E^+ such as $a(9)$ are not cautious consequences of that framework.

H4b (cautious learning changes the root outcome). Once target-wise cautious mode is supported, re-running target a (and symmetrically b) on the **same** frozen sample with ECAI-like folding options but `learning_mode(cautious)` yields an outcome **different** from the brave solve: either `completed_no_solution`, or a solved delta **qualitatively different** from the 11-clause α_2/α_3 choice gadget (e.g. more rote residual structure on the ambiguous $(other, c) = (0, 0)$ cell). The brave choice-gadget solution is **not** expected to reappear unchanged.

H4c (fixture link — non-claim). Interpret any difference against the AND-collider fact that roots have **no** deterministic mechanism over the other observed variables, and that $(B, C) = (0, 0)$ (resp. $(A, C) = (0, 0)$) is label-ambiguous for the root target. H4 tests **learning-mode / entailment semantics** on the target-wise path, not DAG recovery.

7.4 Explicit non-claims for H4

- Not claimed that cautious mode is "better" for causal recovery.
- Not claimed that cautious mode will recover a correct root mechanism rule (none exists).
- Not a ranking of brave vs cautious as general ABALearn policy beyond this controlled probe.
- Not claimed that the probe can be executed with today's target-wise validator as-is.

7.5 Planned investigation (when H4 is started)

Not approved now. When Samuel opens H4, proceed in two stages after resolving the deferred decisions below:

1. **Infra:** implement cautious support for target-wise (new `.pl` + validator/README/tests + distinct `configuration.id`); leave brave `ecai2024` outputs immutable.
2. Optionally cheap **H4a** pre-check: cautious-consequence inspection of the existing brave framework without a full re-learn.

3. **H4b run:** target-wise cautious collection on `n30_seed42` for a/b (optional c).
4. Compare outcomes, deltas, and trace entailment KO points to the locked brave ECAI cells.

7.6 Decisions deferred until H4 begins (flagged; not decided now)

These must be resolved **when H4 is started**, not before H4 is explicitly opened. They are recorded here so the infra prerequisite is not forgotten:

1. **Cautious Prolog filename** — e.g. a twin such as `configs/ecai2024_cautious_config.pl` (illustrative only); do **not** edit `configs/ecai2024_config.pl` in place.
2. **New target-wise** `configuration.id` and output path so brave `ecai2024` collections and hashes remain immutable.
3. **Validator change** — approve relaxing `causal/targetwise/config.py` (`_read_learning_mode`) so `learning_mode(cautious)` is explicitly allowed when selected via that `.pl`; update README/tests accordingly.
4. `check_ic` / `.sol_chk.asp` / **post-run artefact audit** under cautious learning — how to interpret joint existential SAT vs cautious consequences so diagnostics are not misread.
5. **H4a pre-check in scope?** — optional cautious-consequence inspection of the locked brave framework before a full re-learn.
6. **Optional target c control** on the cautious collection, or roots a/b only.

Until those are decided and infra lands, H4 remains **proposed** / **infra-blocked** / **not tested** / **not a claim**.

8 Boundary distinctions to preserve

Keep separate throughout any future probe write-up:

Object	Role
AAMAS strategy / greedy maximal co-occurrence search	procedural learner behaviour (H1/H2)
ECAI nd first-fold / BK serialisation order	procedural learner behaviour (H3)
Brave vs cautious entailment / learning mode	procedural / semantic learner behaviour (H4)
Finite-sample support and label conflicts in candidate body cells	sample information
Mechanism correspondence for c (AND)	evaluator-only interpretation of a non-root
Absence of deterministic root mechanisms	evaluator reference status
Syntactic delta = reference string	coincidence, not automatic causal verdict
Runner `solved` under brave	brave coverage of $E^{\pm}$, not mechanism recovery

Do not treat predictive rules for roots as mechanism recovery.

9 Cross-links

- Investigation record: `experiment.md`
- Mathematical fixture dossier: `fixture_dossier.tex`

- Bucket 3 planning hub: `../M1.3-bucket3-claims.md` (still **no claim**)
- Research decision log: `docs/research/decisions.md` (2026-08-01 H1–H3 entries)
- Fixture pre-run bundle: `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/`
- AAMAS baseline: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/`
- ECAI baseline (H3/H4 motivation): `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/ecai2024/n30_seed42/`
- Target-wise brave-only validator: `causal/targetwise/config.py`
- Entailment implementation: `asp_engine.pl` (`entails/5`)

10 Decisions recorded

Item	Decision	Date
H1 nested-prefix / missing-$(0, 0, 0)$ AAMAS ablation	**Approved** then **run** / **analysed**	2026-08-01 / 2026-08-02
H2 four-variable isolated-D fixture + AAMAS c	**Approved** then **run** / **analysed**	2026-08-01 / 2026-08-02
H2c in first H2 write-up	**In scope** (run; brief note only)	2026-08-01 / 2026-08-02
H3 graph/naming; B 80/20, D 70/30; id `m13_bucket3_binary_bd_and_lead_a`; AAMAS contrast	**Design approved** then **run** / **analysed**	2026-08-01 / 2026-08-02
H4 cautious vs brave on ECAI roots	**Proposed only** (blocked on infra)	—

11 Current status and remaining work

- **H1: complete** (`h1_support_ablation.md`). Not a claim.
- **H2: complete** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains irrelevant d. Not a claim.
- **H3: complete** (`h3_bk_leading_distractor.md`). Lead: ECAI c first-folds to / retains BK-leading a (not BD AND). Not a claim.
- **H4:** deferred until H4 is opened. Infra/config decisions are **flagged** in the H4 section (“Decisions deferred until H4 begins”). No cautious collection until then.
- Still **no** Bucket 3 claim and **no** expanded run matrix without a further Samuel decision.